token at the location of the masked span in the verse, so the sequence became $S = (w_1, \ldots, E_0, \ldots, w_n)$. We then prompted the model to predict the tokens replacing the special E_0 token, computing the cross-entropy loss for the prediction. This pre-trained model, serving as an external scoring mechanism, was independent of our dataset. This setup allows us to verify whether or not our training process impacts the coherence of the model.

Alignment Scores evaluate the alignment of generated words with the required beat rhythm. We employed two metrics:

- *Exact Alignment Accuracy:* This metric checks whether the generated word precisely aligns with the expected beat rhythm, resulting in a binary outcome (0 for non-alignment, 1 for exact alignment).
- *Levenshtein distance:* This metric assesses alignment by calculating the normalized Levenshtein distance between the generated word and the expected beat rhythm. This measure accounts for cases where the alignment may not be exact but is reasonably close, providing additional flexibility in evaluation.

Results

This section explores how well the fine-tuned ByT5 models meet the task's objectives. We evaluate their performance in terms of Coherence and Beat Alignment. In the following section, we discuss the implications of these results for further research.

Coherence: The results in Table 1 show that both models achieved a comparable log perplexity with the baseline model, with the ByT5-CV version slightly ahead with a log-perplexity score of 7.48, compared to 7.494 for the ByT5-B version. The baseline models score between 7.442 and 7.432. These outcomes suggest that the custom fine-tuning process didn't significantly impact the model's semantic coherence, indicating that it adapted well to the rhythmic task without sacrificing fluency. We did not evaluate GPT-4 output for coherence, given that T5 exhibits lower fluency compared to GPT-4.

Alignment Scores: Both of our models showed a high level of beat alignment accuracy. The ByT5-CV model achieved an Exact Alignment Accuracy score of 98.88%, while the ByT5-B variant achieved 98.31%. The Levenshtein distance, which allows for some flexibility, indicated near-perfect scores of 99.83% and 99.63% respectively, suggesting that the models generally aligned with the rhythm, with only minor deviations. This demonstrates that the models understand rhythmic patterns even when they do not exactly match. In contrast, the baseline ByT5 models, including GPT-4 on the 1000 examples, had much lower scores for both exact alignment and Levenshtein distance, indicating that these patterns are challenging to learn without proper training.

These findings confirm that our fine-tuned ByT5 models can generate words that align with beat patterns while maintaining coherence. This sets the stage for further research into generating entire verses or poems with consistent rhythmic structures.

Model	Coherence	Accuracy	Levenshtein
ByT5-Base-CV	7.442	0.3512	0.7670
ByT5-Base-B	**7.432**	0.5575	0.8323
GPT-4	-	0.2550	0.6105
ByT5-CV	7.480	**0.9888**	**0.9983**
ByT5-B	7.494	0.9831	0.9963

Table 1: Performance comparison between our consonant-vowel model, beat model and the baselines. For Coherence, lower is better. For Exact Alignment Accuracy and Levenshtein distance, higher is better.

Conclusion and Future Work

In this study, we investigated the capabilities of ByT5, a byte-level language model, particularly its aptitude for generating words that conform to specific consonant-vowel patterns and rhythmic beats. Our methodology focused on fine-tuning multiple ByT5-based models on a conditional mask-predict objective to reconstruct words with predetermined consonant-vowel patterns and rhythmic beats. The results demonstrated that our models were able to generate words that align with specified patterns while maintaining semantic coherence. Our models showed high rhythmic alignment accuracy indicating their effectiveness in this task without adversely sacrificing the models' fluency.

The models have potential applications in various co-creative frameworks, including songwriting, rap lyric generation, and in the process of rhythmic poetry creation. In future work, we aim to expand the model's capabilities to not only insert individual phrases but also generate complete verses or poems that maintain rhythmic coherence throughout. We will also explore the impact of unstressed syllables on beat strength and the precise locations and durations of beats backed by human evaluation. By advancing these aspects, we intend to enhance the model's utility furthering the integration between poetry and music.

Acknowledgements

We thank the anonymous reviewers for their valuable feedback and members of the Serious Games Research Group at the University of Calgary for their support. We appreciate the scholarship provided by Elsevier's Artificial Intelligence Journal through the ICCC program committee.

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